

FINANCIAL ANALYTICS DASHBOARD USING PYTHON, SQL, AND POWER BI

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Role: Aspiring Data Analyst

Tools Used: Python, Pandas, SQL (SQLite), Power BI, Excel

1. Introduction

In today's data-driven economy, financial analytics has become the backbone of strategic business management. Organizations generate vast amounts of transactional data daily, yet raw data alone holds little value. The true competitive advantage lies in transforming this data into actionable intelligence that drives profitability, optimizes operational efficiency, and mitigates financial risks.

This project simulates a real-world Business Intelligence (BI) engagement. It presents a complete Financial Analytics Dashboard developed using Python, SQL, and Power BI. The project analyzes a supermarket sales dataset to evaluate financial performance across multiple branches. By tracking key metrics such as revenue, expenses, and profit margins, the dashboard enables stakeholders to quickly identify trends, pinpoint underperforming areas, and make data-backed decisions.

The project strictly follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, covering the entire analytics lifecycle: Business Understanding, Data Understanding, Data Preparation, Modeling (EDA), Evaluation, and Deployment.

2. Project Objectives

The primary objectives of this project are structured to solve specific business problems:

- **Data Integrity:** To clean and preprocess raw financial transaction data, ensuring 100% accuracy for reporting by handling missing values, duplicates, and formatting inconsistencies.
 - **KPI Tracking:** To calculate and monitor core financial health indicators including Total Revenue, Total Expenses, Net Profit, and Profit Margin.
 - **Trend Analysis:** To analyze monthly financial performance to identify seasonal patterns or growth trajectories.
 - **Product Strategy:** To identify high-performing product categories that generate the most revenue and profit, providing data for inventory and marketing strategies.
 - **Geographical Performance:** To compare profitability across different cities and branches to determine which regions require operational support or resource allocation.
 - **Dashboard Deployment:** To create an interactive, user-friendly Power BI dashboard that allows non-technical business users to filter data dynamically and derive insights independently.
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3. Dataset Description

Dataset Name

Supermarket Sales Dataset

Source

Kaggle (Public Domain)

Dataset Summary

The dataset contains historical transaction records from three supermarket branches located in different cities within Myanmar. It simulates the point-of-sale (POS) data typically available in corporate finance departments.

Data Statistics

- **Total Records (Rows):** 1000
- **Total Columns:** 17
- **Time Period:** January 2019 – March 2019 (3 months)

Main Attributes (Data Dictionary)

Column Name	Description	Data Type
Invoice ID	Unique identifier for each transaction (e.g., 750-67-8428)	String
Branch	Branch code (A, B, or C)	String
City	Location of the branch (Yangon, Mandalay, Naypyitaw)	String
Customer Type	Member or Normal customer	String
Gender	Gender of the customer	String
Product Line	Category of the product sold (e.g., 'Food and Beverages', 'Electronic Accessories')	String
Unit Price	Price per unit of the product	Float
Quantity	Number of units purchased	Integer
Tax (5%)	5% tax applied to the purchase	Float
Total (Revenue)	Pre-tax total amount (Unit Price * Quantity)	Float
Cogs (Expense)	Cost of Goods Sold (the cost incurred by the supermarket to acquire the product)	Float
Gross Income (Profit)	Pre-tax profit (Total - Cogs)	Float
Date	Date of the transaction	Date
Time	Time of the transaction	Time
Payment	Method used (Cash, Credit Card, Ewallet)	String
Rating	Customer satisfaction rating (1 to 10)	Float

4. Data Cleaning Process (Python - Pandas)

Raw data is often unusable. To ensure the integrity of the financial calculations, I used **Python** and the **Pandas** library to perform rigorous data preprocessing.

Step-by-Step Cleaning Procedure:

1. **Loading and Inspection:** Loaded supermarket_sales.csv into a DataFrame and ran .info() and .describe() to understand the data structure.
2. **Column Standardization:** Renamed columns for consistency (e.g., converting 'Total' to 'Revenue', 'Cogs' to 'Expense', 'Gross Income' to 'Profit') to avoid confusion during SQL querying.
3. **Duplicate Removal:** Checked for duplicate Invoice ID entries. Although the dataset had none, the code df.drop_duplicates(subset=['Invoice ID'], inplace=True) was implemented to handle this generically.

4. **Type Conversion:** Ensured Unit Price, Quantity, Tax, Revenue, Expense, and Profit were converted to float or int using `pd.to_numeric()`.
 5. **Date Parsing:** Converted the Date column from a standard string to a Pandas datetime object (`df['Date'] = pd.to_datetime(df['Date'])`). This was critical to enable monthly trend analysis and time-series visualizations.
 6. **Missing Values:** Checked for null values. No significant missing data was found, ensuring no imputation was required which could skew financial results.
 7. **Saving Clean Data:** Exported the polished dataset as `cleaned_supermarket_data.csv` to serve as the single source of truth for both SQL analysis and Power BI.
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5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Matplotlib and Seaborn to uncover underlying patterns, anomalies, and relationships before building the dashboard.

Key Visualizations Created:

- **Correlation Heatmap:**
 - *Implementation:* Plotted to examine multicollinearity between numerical variables (Revenue, Expense, Profit, Quantity, Tax, Rating).
 - *Insight:* A strong positive correlation (≈ 0.99) was observed between Revenue and Expense. This is expected, as higher sales volumes typically incur higher COGS. However, Profit showed a moderate correlation with Quantity, suggesting volume is a key driver of bottom-line growth.
 - **Monthly Revenue Trend:**
 - *Implementation:* Aggregated revenue by month and plotted a line chart. A rolling average (3-period moving average) was overlaid to smooth out day-to-day volatility.
 - *Insight:* Revenue peaked in mid-February, likely due to a seasonal promotion or holiday, while March showed a slight dip, indicating potential post-holiday demand shrinkage.
 - **Expense Boxplot (Outlier Detection):**
 - *Implementation:* A boxplot was used to visualize the distribution of Expenses.
 - *Insight:* A few high-expense outliers were identified. Upon investigation, these corresponded to bulk purchase transactions in the "Home and Lifestyle" product line. This is legitimate business behavior and not data entry error.
 - **Revenue Distribution:**
 - *Implementation:* A histogram with a Kernel Density Estimate (KDE) curve.
 - *Insight:* The distribution is right-skewed, meaning the majority of transactions are small (around \$10-\$30), but a long tail of high-ticket transactions drives the majority of total revenue.
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6. SQL Analysis (SQLite)

To practice data extraction for specific business questions, I imported the cleaned dataset into an **SQLite** database. SQL allowed me to aggregate data efficiently before loading it into Power BI.

Executed SQL Queries:

1. **Monthly Revenue Trend:**

```
SELECT strftime('%m', Date) AS Month,  
       ROUND(SUM(Revenue), 2) AS Total_Revenue  
FROM Supermarket_Sales  
GROUP BY Month  
ORDER BY Month;
```

2. Revenue by Product Line:

```
SELECT Product_Line,  
       SUM(Quantity) AS Units_Sold,  
       ROUND(SUM(Revenue), 2) AS Revenue_Generated  
FROM Supermarket_Sales  
GROUP BY Product_Line  
ORDER BY Revenue_Generated DESC;
```

3. Profit by City:

```
SELECT City,  
       ROUND(SUM(Profit), 2) AS Total_Profit  
FROM Supermarket_Sales  
GROUP BY City  
ORDER BY Total_Profit DESC;
```

4. Payment Method Popularity:

```
SELECT Payment,  
       COUNT(*) AS Transaction_Count,  
       ROUND(SUM(Revenue), 2) AS Total_Revenue  
FROM Supermarket_Sales  
GROUP BY Payment;
```

5. Profit Margin Calculation (Aggregated):

```
SELECT ROUND(((SUM(Profit) / SUM(Revenue)) * 100), 2) AS Company_Profit_Margin  
FROM Supermarket_Sales;
```

(Result: The company achieved an approximate 4.76% profit margin, aligning with standard retail grocery margins).

7. Power BI Dashboard Development

The core deliverable of this project is an interactive Power BI dashboard. It is designed with a top-down analytical structure: summary first, details second.

A. Dashboard Layout & KPI Cards

At the top of the canvas, I placed four large KPI (Key Performance Indicator) cards:

- **Total Revenue:** Displayed in the local currency format.

- **Total Expenses (COGS):** Highlighted in a neutral blue.
- **Total Net Profit:** Highlighted in green.
- **Profit Margin (%):** Displayed as a percentage.
- *(DAX Measure for Margin: Profit Margin = DIVIDE(SUM('Sales'[Profit]), SUM('Sales'[Revenue]), 0) * 100)*

B. Core Visualizations

1. **Line Chart (Monthly Revenue Trend):** Placed in the top-left of the main chart area. Used to quickly identify growth/decline patterns over the 3-month window.
2. **Stacked Bar Chart (Revenue by Product Line):** Sorted descending to highlight the top revenue generators ("Food and Beverages" and "Electronic Accessories").
3. **Clustered Column Chart (Profit by City):** Used to geographically compare branch performance.
4. **Donut Chart (Payment Method):** Shows the market share of Cash, Credit Card, and Ewallet transactions.
5. **Scatter Plot (Quantity vs. Revenue):** Used to detect high-volume, low-revenue items versus low-volume, high-revenue items.

C. Interactive Features (Filters/Slicers)

To empower users, I added global filters in a side panel:

- **City Slicer:** Dropdown to view specific branches (Yangon, Mandalay, Naypyitaw).
- **Product Line Slicer:** Multi-select to analyze specific categories.
- **Customer Type Slicer:** Toggle between "Member" and "Normal" to see loyalty program impact.
- **Date Slicer:** A slider to narrow down to specific date ranges.

8. Technologies Used

Technology	Purpose
Python (Pandas)	Loading, cleaning, aggregating, and reshaping the raw CSV data.
Python (Matplotlib/Seaborn)	Conducting exploratory data analysis and generating statistical plots.
SQL (SQLite)	Writing complex queries to validate business logic and aggregate data.
Microsoft Power BI	Connecting to the data source, creating relationships, writing DAX measures, and designing the interactive dashboard interface.
Microsoft Excel	Initial data inspection (viewing CSV in grid format) and quick pivot table validation.
Jupyter Notebook	An interactive environment used for executing the Python code step-by-step.

9. Key Business Insights & Recommendations

Based on the data analysis and dashboard interactions, the following actionable insights were drawn:

1. **Product Focus Strategy:** *Food and Beverages* and *Electronic Accessories* account for nearly 45% of total revenue. **Recommendation:** Increase shelf space and marketing budget for these categories during peak months (January/February) to maximize returns.

2. **Branch Performance Gap:** The city of *Naypyitaw* (Branch C) is significantly underperforming in profit compared to Yangon and Mandalay. **Recommendation:** Investigate local management, supply chain costs, or customer footfall in Naypyitaw to identify if this is a staffing issue or a market size issue.
 3. **Payment Flexibility:** Credit Card and Ewallet transactions combined account for over 70% of revenue. **Recommendation:** The supermarket should invest in modern POS systems that expedite cashless transactions and consider running cashback promotions to boost customer loyalty.
 4. **Seasonal Fluctuations:** Revenue and Expenses moved in perfect sync, with January showing the lowest dip. **Recommendation:** The company should launch a "New Year" sales campaign in early January to stimulate demand during historically slow periods.
 5. **Customer Loyalty:** Member customers generated a higher average profit margin than Normal customers. **Recommendation:** Broaden the membership program incentives to convert more walk-in customers into members.
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10. Project Workflow

The project followed a linear yet iterative pipeline to ensure accuracy:

1. **Data Acquisition:** Downloaded raw CSV from Kaggle.
 2. **Data Processing (Python):** Cleaned and transformed data using Pandas; exported to `cleaned_data.csv`.
 3. **Data Storage/Query (SQL):** Loaded the clean CSV into an SQLite database; wrote queries to validate aggregations.
 4. **Exploratory Analysis (Python):** Visualized distribution and correlations to find initial trends.
 5. **Dashboard Development (Power BI):** Connected to the CSV, created relationships, built DAX measures, and designed the UI layout.
 6. **Testing & Validation:** Cross-referenced Power BI totals against SQL outputs to ensure zero mathematical errors.
 7. **Publishing:** Deployed the dashboard to Power BI Service and documented insights.
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11. Challenges Faced and Resolutions

Throughout this project, I encountered several practical roadblocks that strengthened my problem-solving skills:

- **Challenge: Column Name Inconsistency.** The original CSV had columns named `cogs` and `gross income`.
 - *Resolution:* Standardized these in Python using `df.rename(columns={'cogs': 'Expense', 'gross income': 'Profit'})` to avoid writing complex SQL aliases every time.
- **Challenge: Date Format Interpretation.** Power BI initially read the Date column as text.
 - *Resolution:* Ensured the Python script correctly parsed the date and exported it in a strict YYYY-MM-DD format, allowing Power BI to automatically recognize it as a Date hierarchy.
- **Challenge: DAX Measure for Profit Margin.**
 - Simply dividing `SUM(Profit)` by `SUM(Revenue)` yields a weighted average, but I initially tried to divide row-by-row, giving incorrect aggregated totals.

- *Resolution:* Used the DIVIDE DAX function correctly, which handles division by zero gracefully and aggregates properly.
 - **Challenge: Dashboard Clutter.** Initially, I used too many visuals on one page.
 - *Resolution:* I applied a "Gestalt principle" by grouping related visuals (Expenses & Revenue) together and adding a clean background grid, reducing visual noise.
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12. Future Improvements

While the current dashboard is fully functional, future iterations could include:

- **Machine Learning Forecasting:** Integrating Python's Prophet or ARIMA models to forecast the next 6 months of revenue based on historical trends, allowing management to plan inventory proactively.
 - **Customer 360 Segmentation:** Using K-Means Clustering to segment customers based on spending habits, enabling hyper-personalized marketing.
 - **Real-Time Integration:** Connecting Power BI to a live SQL database (Azure SQL/PostgreSQL) instead of a static CSV, ensuring the dashboard updates automatically with daily sales.
 - **Automated ETL:** Using Apache Airflow to automate the data cleaning Python script, removing the need for manual file processing.
 - **Web App Deployment:** Converting the Python EDA into an interactive Streamlit application that allows non-Power BI users to explore the raw data analytics.
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13. Conclusion

This project successfully demonstrates an end-to-end financial analytics workflow using the modern data analytics stack: Python for heavy lifting, SQL for extraction, and Power BI for visualization. By transforming raw, messy transaction data into a polished, interactive dashboard, this project provides a clear solution for executive decision-making.

The dashboard effectively enables stakeholders to answer critical business questions such as "Which city is losing money?" and "Which products should we promote?" in seconds. This project not only validates my technical proficiency in Python, SQL, and Power BI but also reinforces my understanding of business finance and data storytelling. It stands as a solid foundation for a professional portfolio, simulating the real-world responsibilities of a data analyst in a corporate or retail finance environment.

14. References

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